

# A NET-LEVEL GEODESIC LIPSCHITZ-FREE SUBCASE OF QUESTION 2.3 IN ARXIV:2210.13306

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ABSTRACT. Ciaci asked whether every locally almost square Banach space satisfies the weakly-null-net conclusion proved in his paper for positive locally almost square Banach lattices. We prove a partial positive result: if  $M$  is a geodesic metric space, then the Lipschitz-free space  $\mathcal{F}(M)$  satisfies that net conclusion at every point of its unit sphere. The proof combines the finite-support geodesic theorem of Kaasik–Veeorg with a simple density lemma showing that pointwise weakly-null square sequences on a dense subset of the sphere promote to weakly-null square nets on the whole sphere.

## 1. SOURCE QUESTION

In [1], a Banach space  $X$  is called locally almost square (LASQ) if, for every  $y \in S_X$  and every  $\varepsilon > 0$ , there exists  $x \in S_X$  such that

$$\|y + x\| \leq 1 + \varepsilon \quad \text{and} \quad \|y - x\| \leq 1 + \varepsilon.$$

Ciaci proves that a positive LASQ Banach lattice has the following stronger net conclusion: for every  $y \in S_X$ , there is a weakly null net  $(x_\alpha)$  such that

$$\|x_\alpha\| \rightarrow 1, \quad \|y + x_\alpha\| \rightarrow 1, \quad \|y - x_\alpha\| \rightarrow 1.$$

Question 2.3 of [1] asks whether the same conclusion follows from LASQ alone, without the lattice and positivity hypotheses.

## 2. IDEA OF THE PROOF

The only new mechanism is that nets are stable under sphere-density in a way that sequences need not be. Suppose a dense set  $D \subset S_X$  consists of points at which weakly-null square sequences already exist. Given a target  $y$ , a weak neighborhood  $U$  of zero, and a tolerance  $\varepsilon$ , choose  $z \in D$  close to  $y$ . Then move far enough along the sequence attached to  $z$  so that the chosen vector lies in  $U$  and is almost square with  $z$ . It is automatically almost square with  $y$ , up to the small error  $\|y - z\|$ . Directing these choices by weak neighborhoods and tolerances gives the desired weakly null net.

Kaasik and Veeorg prove in [2] that for a geodesic metric space  $M$ , every finite-support  $y \in S_{\mathcal{F}(M)}$  has such a weakly-null square sequence. Since finite-support elements are dense in  $\mathcal{F}(M)$ , the density lemma promotes their result to all  $y \in S_{\mathcal{F}(M)}$ , at the net level.

### 3. THE DENSITY LEMMA

**Definition 1.** Let  $X$  be a Banach space. A point  $y \in S_X$  has the *weakly-null-net square property* if there is a weakly null net  $(x_\alpha) \subset B_X$  such that

$$\|x_\alpha\| \rightarrow 1, \quad \|y + x_\alpha\| \rightarrow 1, \quad \|y - x_\alpha\| \rightarrow 1.$$

**Lemma 1** (Density promotion). *Let  $D \subset S_X$  be norm dense. Assume that every  $z \in D$  has a weakly null sequence  $(x_n^z) \subset B_X$  satisfying*

$$\|x_n^z\| \rightarrow 1, \quad \|z + x_n^z\| \rightarrow 1, \quad \|z - x_n^z\| \rightarrow 1.$$

*Then every  $y \in S_X$  has the weakly-null-net square property.*

*Proof.* Fix  $y \in S_X$ . It is enough to show the following local selection property: for every  $\varepsilon > 0$  and every weak neighborhood  $U$  of zero, there exists  $x \in U \cap B_X$  such that

$$\|x\| > 1 - \varepsilon, \quad \|y + x\| < 1 + \varepsilon, \quad \|y - x\| < 1 + \varepsilon.$$

Indeed, after choosing one such  $x_{\varepsilon,U}$  for every pair  $(\varepsilon, U)$ , ordered by decreasing  $\varepsilon$  and shrinking neighborhoods, the resulting net is weakly null and has the required norm limits.

Let  $\varepsilon > 0$  and let  $U$  be a weak neighborhood of zero. Choose  $\eta > 0$  with  $3\eta < \varepsilon$ , and choose  $z \in D$  such that  $\|y - z\| < \eta$ . Since the sequence attached to  $z$  is weakly null and has the square limits at  $z$ , there is  $n$  such that  $x := x_n^z \in U$ ,  $\|x\| > 1 - \eta$ , and

$$\|z + x\| < 1 + \eta, \quad \|z - x\| < 1 + \eta.$$

Therefore

$$\|y \pm x\| \leq \|y - z\| + \|z \pm x\| < \eta + (1 + \eta) < 1 + \varepsilon$$

and  $\|x\| > 1 - \varepsilon$ . This proves the local selection property.

Finally, the two upper estimates force the lower limits of  $\|y \pm x_{\varepsilon,U}\|$  to be 1: if both upper bounds are at most  $1 + \varepsilon$ , then

$$2 = \|(y + x_{\varepsilon,U}) + (y - x_{\varepsilon,U})\| \leq \|y + x_{\varepsilon,U}\| + \|y - x_{\varepsilon,U}\|,$$

so each of the two norms is at least  $1 - \varepsilon$  once the other is at most  $1 + \varepsilon$ .  $\square$

### 4. APPLICATION TO GEODESIC LIPSCHITZ-FREE SPACES

**Theorem 1.** *Let  $M$  be a geodesic metric space. Then for every  $y \in S_{\mathcal{F}(M)}$  there exists a weakly null net  $(\mu_\alpha) \subset B_{\mathcal{F}(M)}$  such that*

$$\|\mu_\alpha\| \rightarrow 1, \quad \|y + \mu_\alpha\| \rightarrow 1, \quad \|y - \mu_\alpha\| \rightarrow 1.$$

*Proof.* Let

$$D = \{z \in S_{\mathcal{F}(M)} : z \text{ has finite support}\}.$$

The linear span of the evaluation vectors  $\delta_p$ ,  $p \in M$ , is dense in  $\mathcal{F}(M)$ . Hence  $D$  is norm dense in  $S_{\mathcal{F}(M)}$ .

By [2, Proposition in Section 3], if  $M$  is geodesic and  $z \in S_{\mathcal{F}(M)}$  has finite support, then there is a weakly null sequence  $(\mu_n) \subset B_{\mathcal{F}(M)}$  such that

$$\|\mu_n\| \rightarrow 1, \quad \|z + \mu_n\| \rightarrow 1, \quad \|z - \mu_n\| \rightarrow 1.$$

Thus every point of the dense set  $D$  satisfies the hypothesis of Lemma 1. Applying that lemma with  $X = \mathcal{F}(M)$  gives the desired weakly null net for every  $y \in S_{\mathcal{F}(M)}$ .  $\square$

**Corollary 1.** *In the class of Lipschitz-free spaces over geodesic metric spaces, Question 2.3 of [1] has a positive answer.*

## 5. LIMITATIONS AND VERIFICATION NOTES

This packet does not prove the full arbitrary-LASQ statement asked in [1]. It also does not settle the length but non-geodesic Lipschitz-free examples from [2]; those examples separate LASQ from WASQ, and the sequence obstruction in that paper does not by itself rule out weakly null nets.

The proof above uses only:

- (1) the standard density of finite-support elements in a Lipschitz-free space;
- (2) the finite-support geodesic weakly-null sequence theorem from [2]; and
- (3) the elementary density-promotion Lemma 1.

No computational verification is required.

## REFERENCES

- [1] S. Ciaci, *Locally almost square Banach lattices*, arXiv:2210.13306, 2022.
- [2] J. K. Kaasik and T. Veeorg, *Weakly almost square Lipschitz-free spaces*, arXiv:2210.09158, 2022.